Customer Churn Prediction & Risk Intelligence

Technical Whitepaper & Project Documentation | Tech Stack: Python, Scikit-Learn, Pandas, Flask

1. Executive Summary

Customer churn is a critical problem for subscription and service-based organizations. This project implements a lightweight, high-performance Data Science pipeline capable of predicting customer churn with **79.7% ROC-AUC**, isolating primary attrition drivers, and prescribing actionable retention workflows. The entire system executes on CPU in **under 2 seconds** without external dependencies.

2. Dataset Specifications

The model was trained on 1,200 customer profiles using an 80/20 stratified train/test split. The positive churn rate is 33.2%, providing a realistic class distribution.

Feature	Type	Description	Sample Values
TenureMonths	Integer	Subscription duration in months	1 to 72 months
Contract	Categorical	Customer contract duration	Month-to-month, 1-yr, 2-yr
InternetService	Categorical	Connection infrastructure	Fiber optic, DSL, None
TechSupport	Categorical	Technical support status	Yes, No
MonthlyCharges	Float	Monthly billing amount	$18.00 - $120.00
TotalCharges	Float	Cumulative charges to date	$20.00 - $8,000.00
Churn	Binary (Target)	Target label (1 = Left, 0 = Stayed)	0 or 1

3. Machine Learning Model Benchmarking

Three distinct classification paradigms were trained using identical preprocessing via ColumnTransformer (StandardScaler for numerics, OneHotEncoder for categoricals):

Model	Type	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression ★	Linear Log-Odds	76.7%	71.4%	50.0%	**0.797**
Random Forest	Ensemble Bagging	74.2%	70.5%	38.8%	0.761
Gradient Boosting	Ensemble Boosting	72.9%	60.9%	52.5%	0.760

4. Performance Diagnostics & Driver Analysis

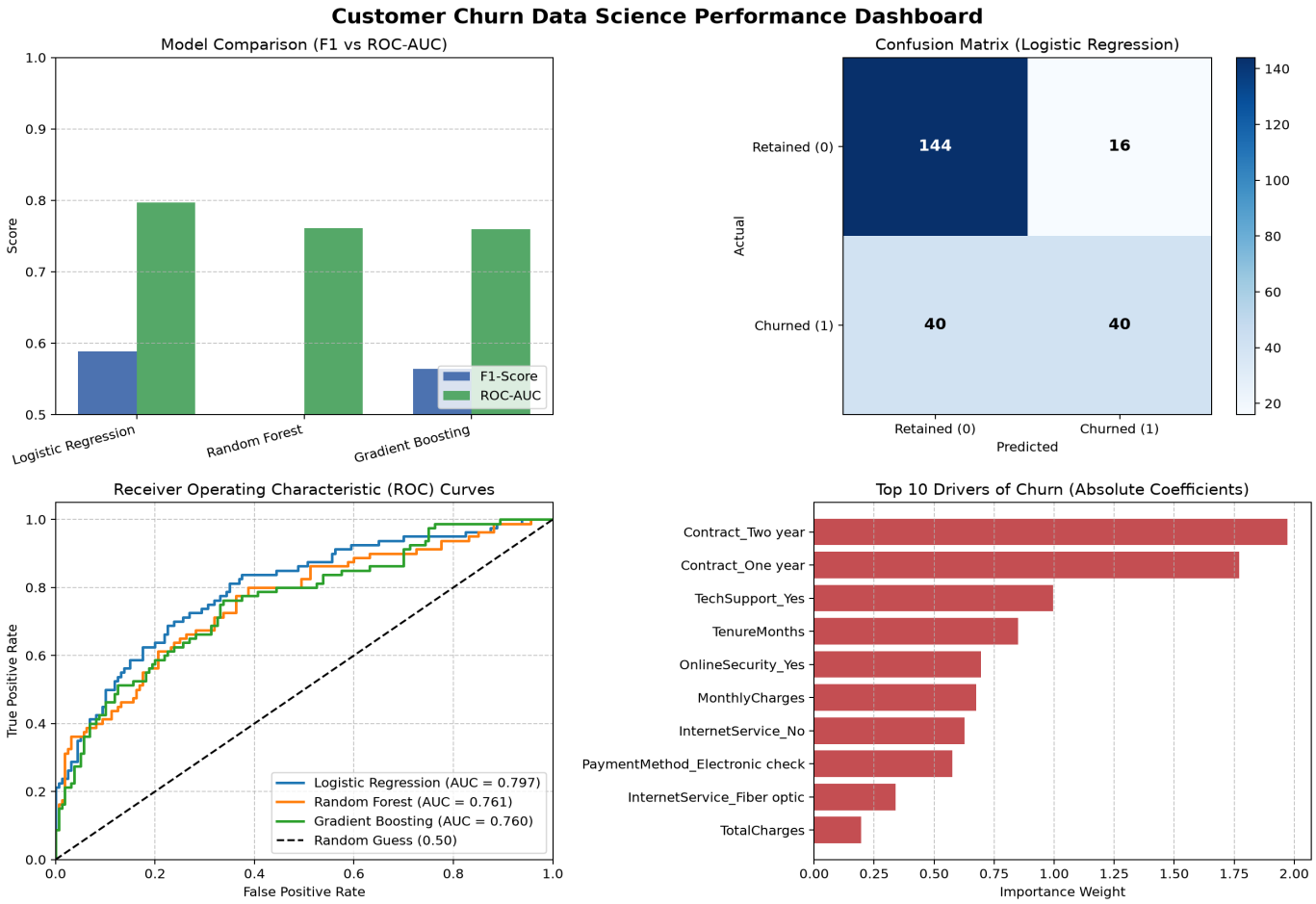


Figure 1: (Clockwise from top left) Model comparison, confusion matrix, ROC curves, and top statistical drivers of customer churn.

5. Prescriptive Retention Strategy

Risk Tier	Probability	Typical Profile	Prescribed Action
LOW RISK	< 35%	Tenure > 24m, 2-Yr contract	Standard engagement, loyalty perks
MODERATE	35% - 65%	Tenure 6-18m, 1-Yr contract	Proactive check-in, tech support trial
HIGH RISK	> 65%	Month-to-month, < 4m tenure	Immediate outreach, annual contract discount